

Cross-validation

STAT 4710/5710

September 16, 2025

Where we are



Unit 1: R for data mining

Unit 2: Prediction foundations

Unit 3: Regression-based methods

Unit 4: Tree-based methods

Unit 5: Deep learning

Lecture 1: Model complexity

Lecture 2: Bias-variance trade-off

Lecture 3: Cross-validation

Lecture 4: Classification

Lecture 5: Unit review and quiz in class

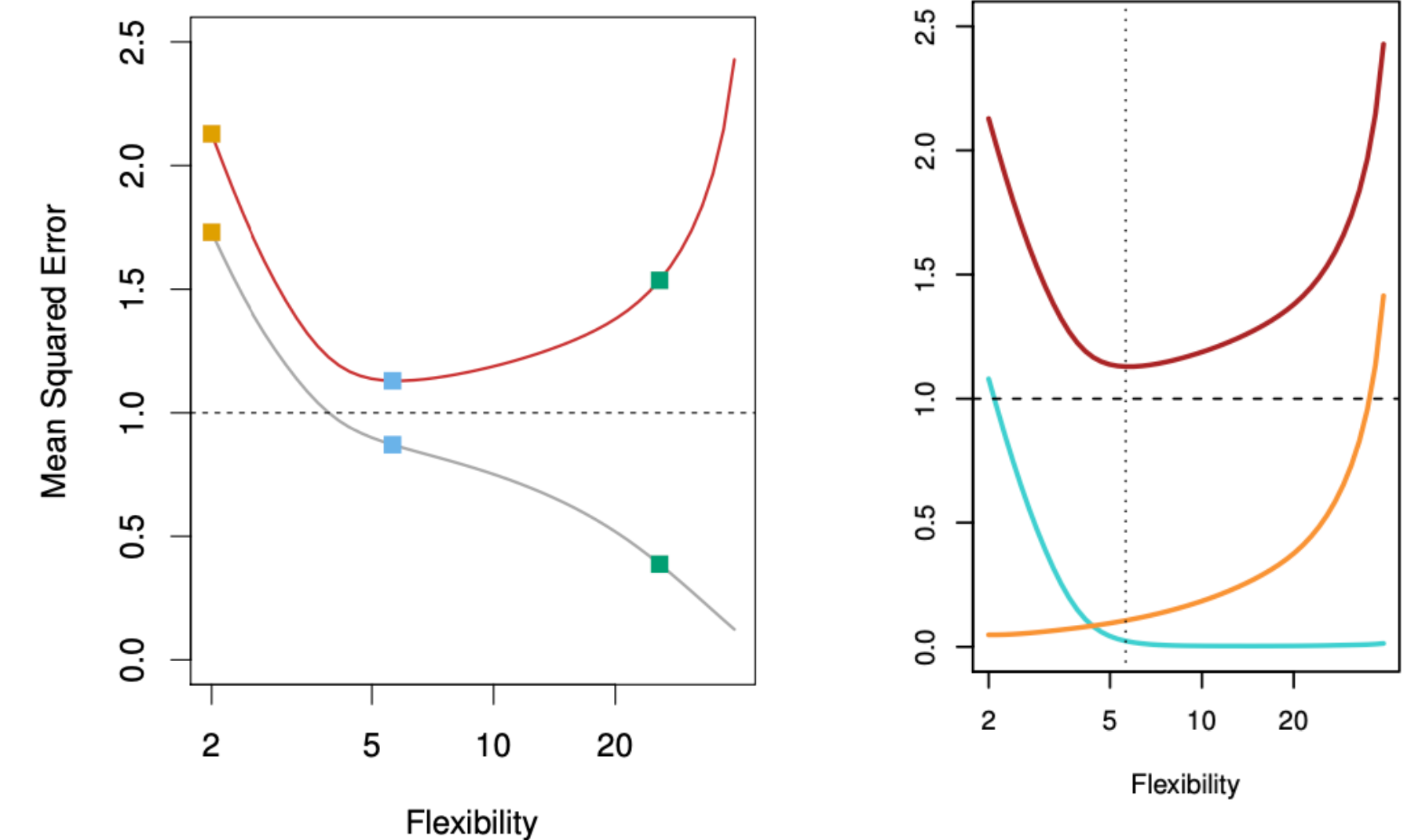
Recap on last 2 lectures

In Lecture 1, we learned

- Notion of model complexity, degree of freedom
- How training error and test error changes over model complexity

In Lecture 2, we learned

- ETE = estimation error + irreducible error
- Estimation error = $\text{bias}^2 + \text{variance}$
- How do model parameters affect these elements?

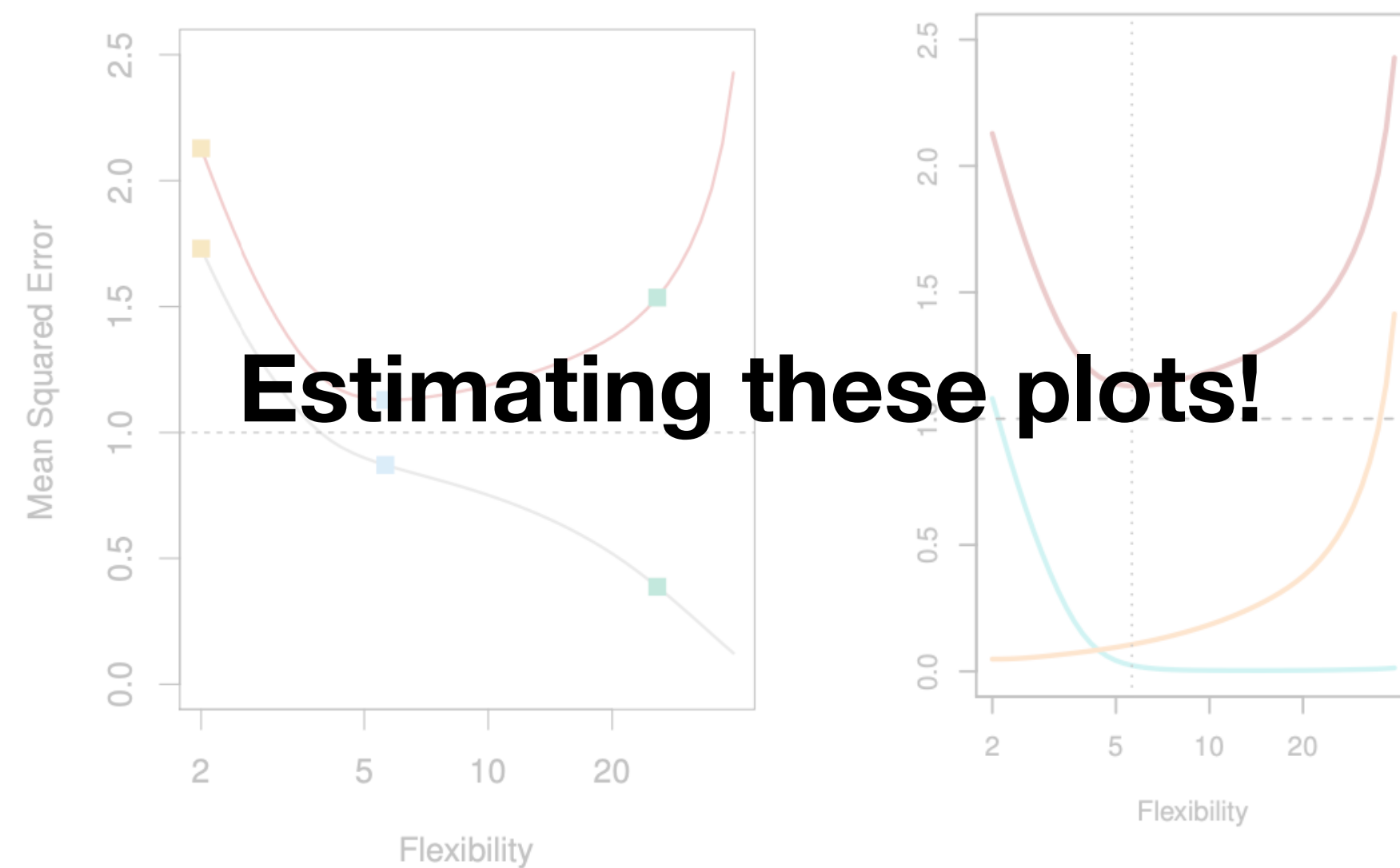


Colored curves are unknown
in a real data mining task!

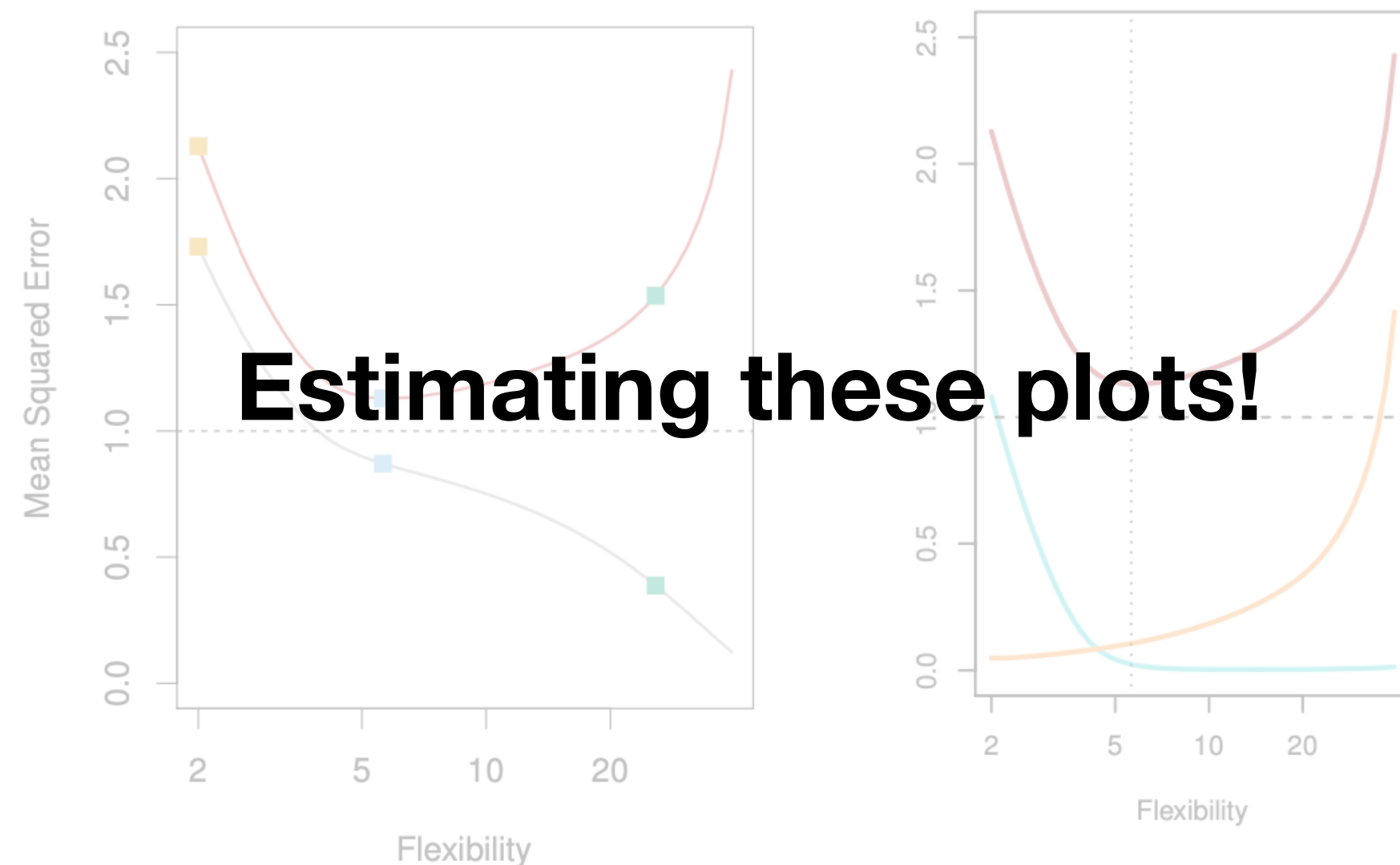
This lecture: From understanding to practice

How do we know where we are, and decide where to stay?

Navigating the bias-variance tradeoff in practice

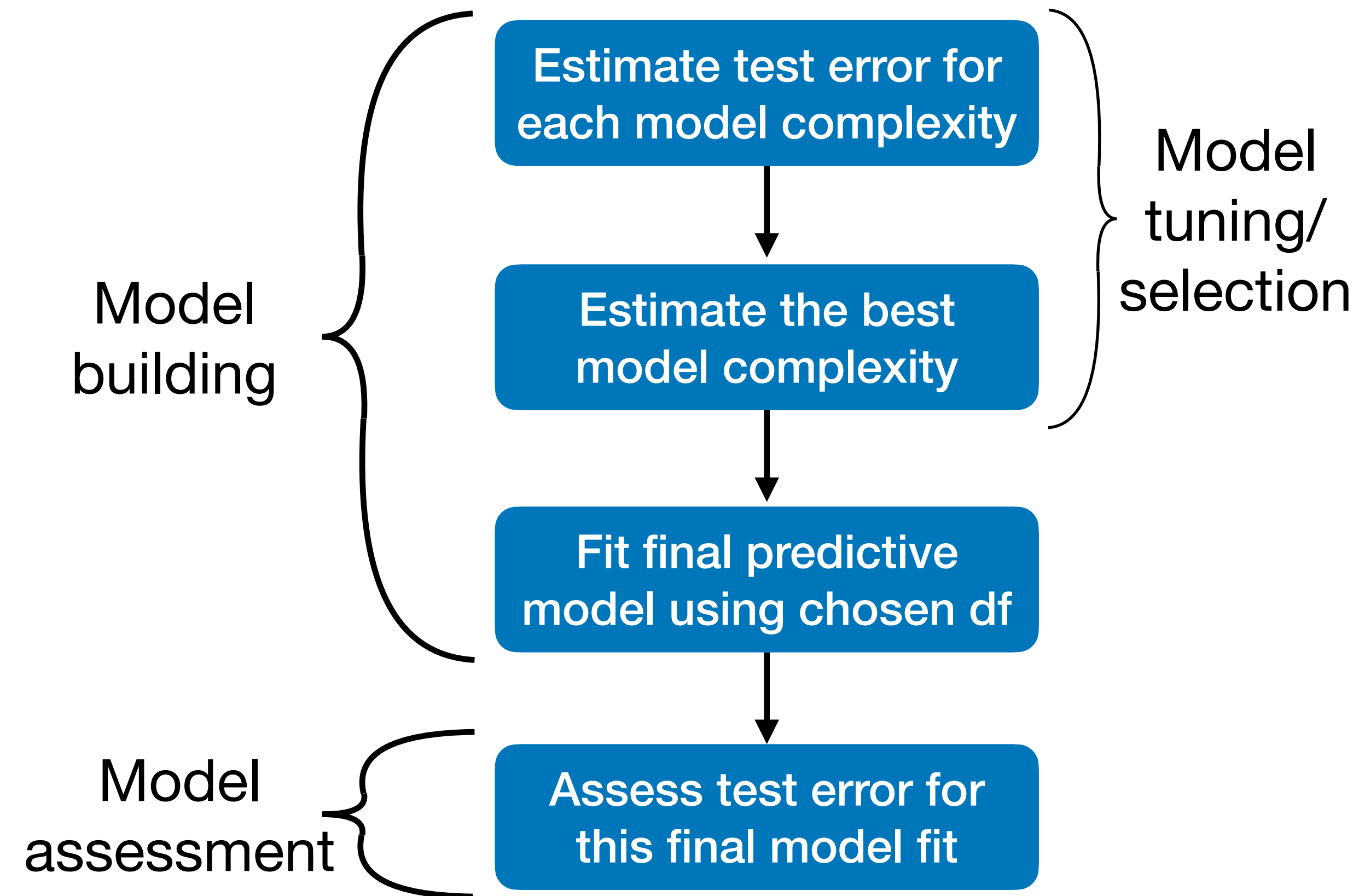


Navigating the bias-variance tradeoff in practice

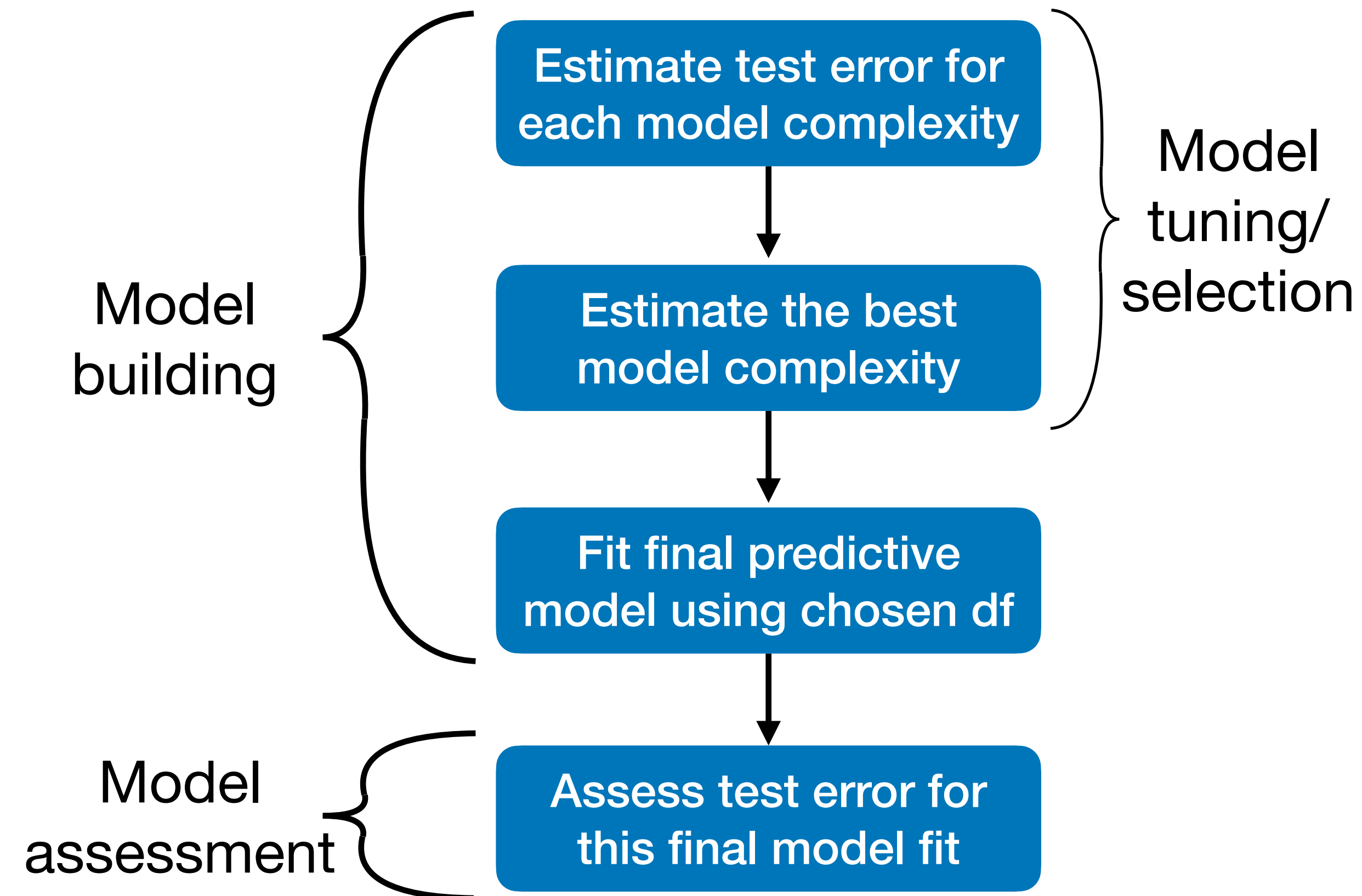
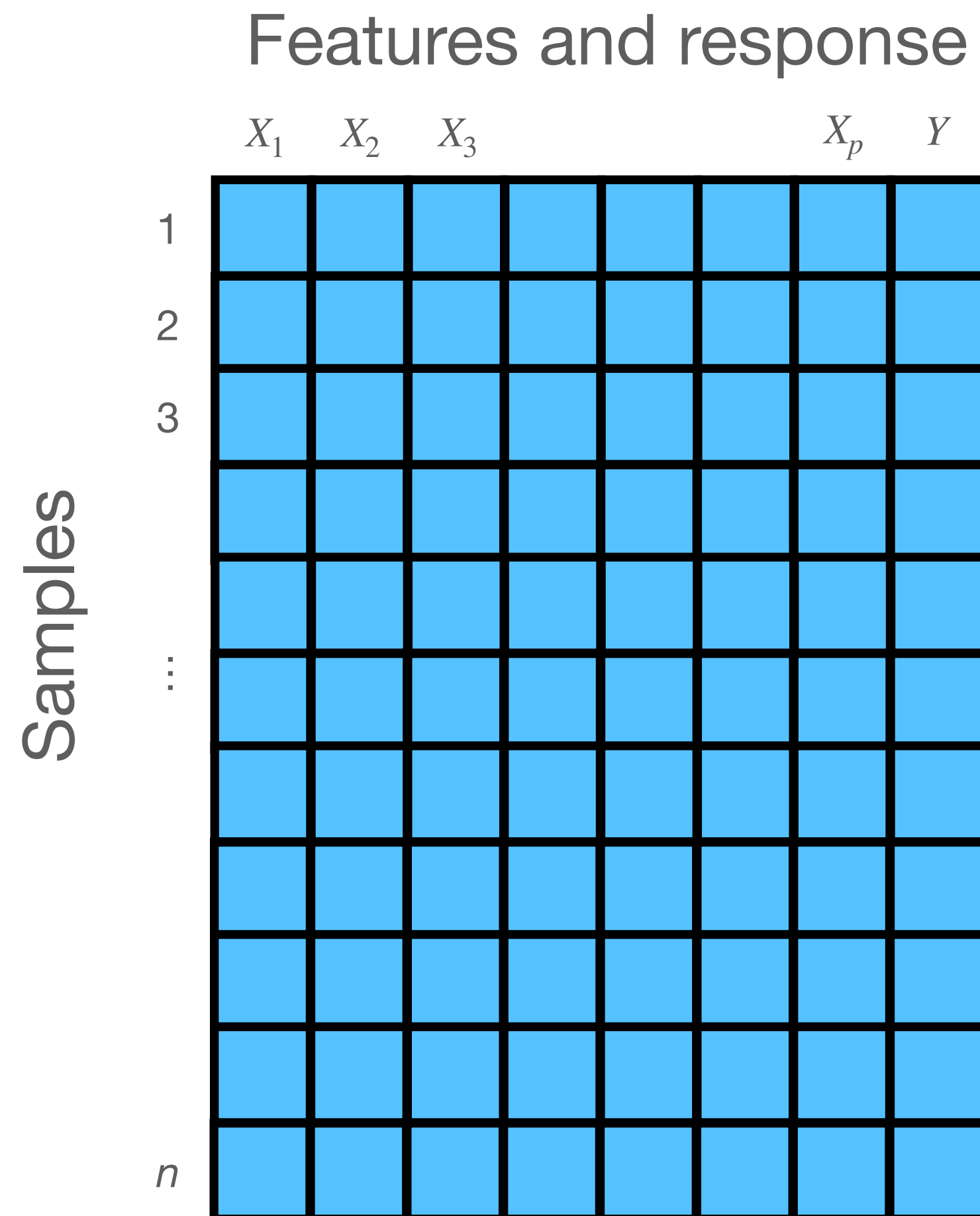


At the end of the day,

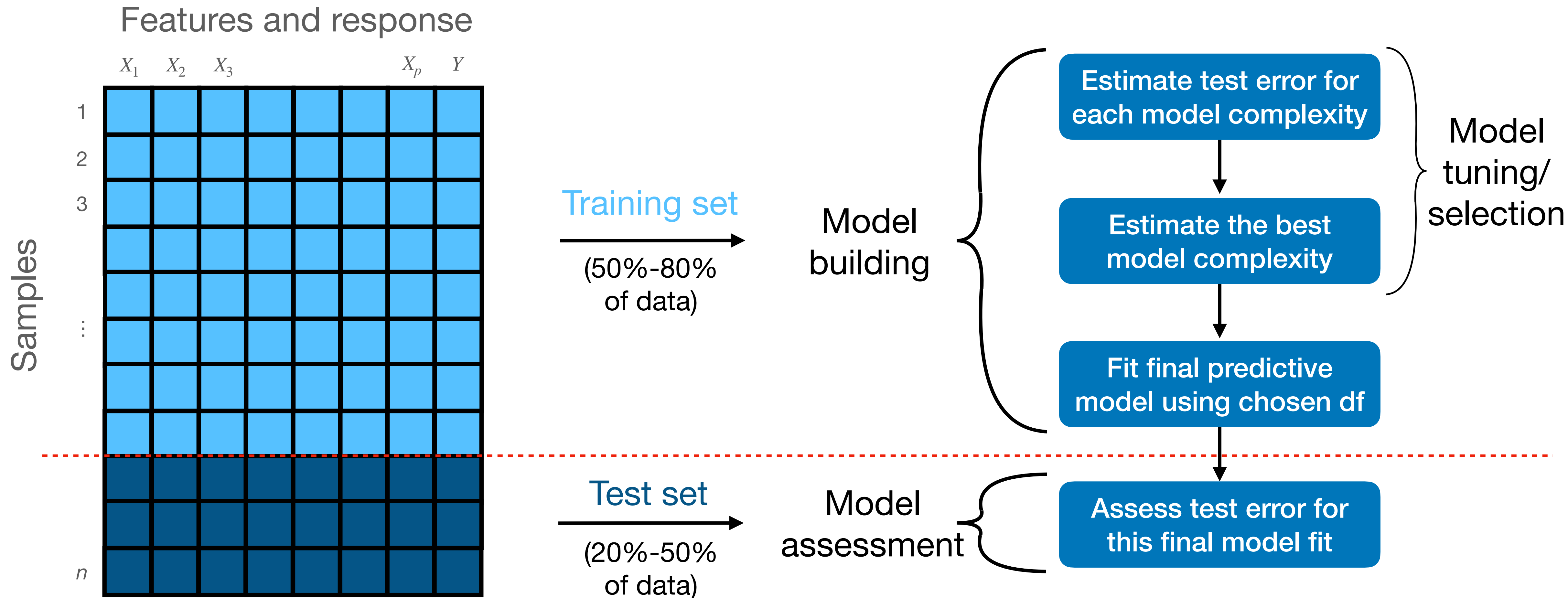
- Have a model to use (building)
- Have a sense of how good it is (assessment)



Separating data for model building and assessment

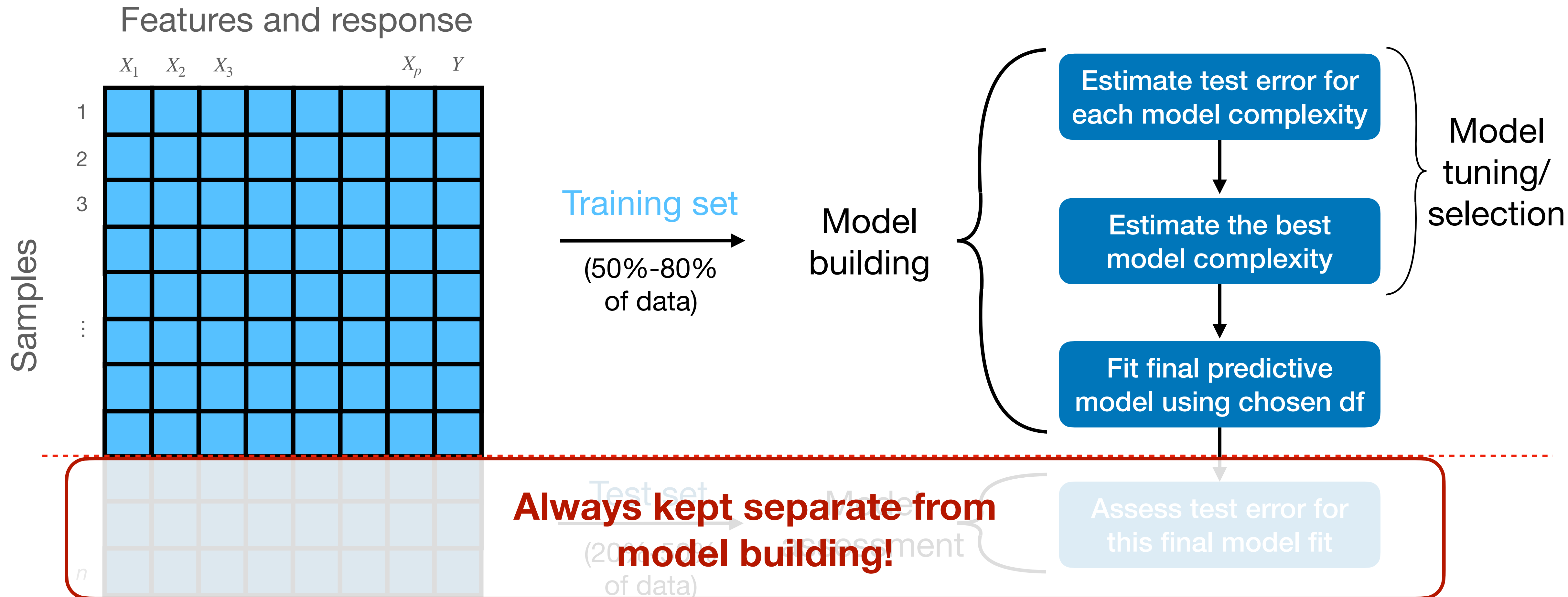


Separating data for model building and assessment



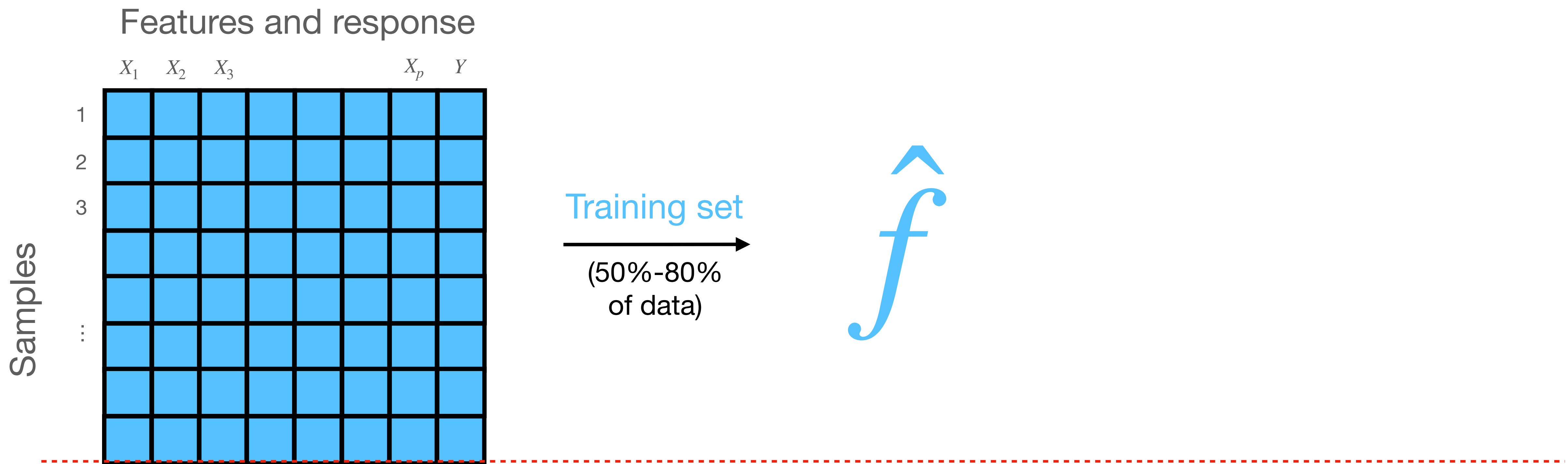
More samples for training $\rightarrow$ better fitted model;
More samples for testing $\rightarrow$ better estimate of test error.

Separating data for model building and assessment

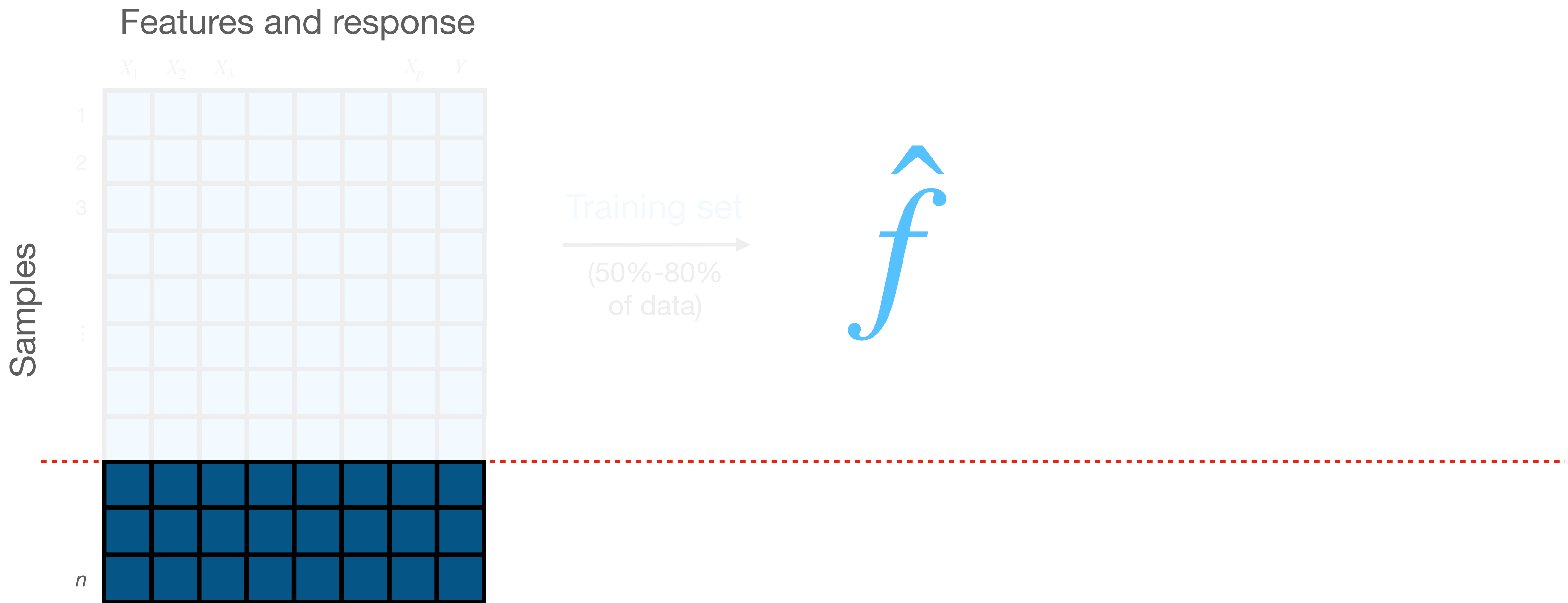


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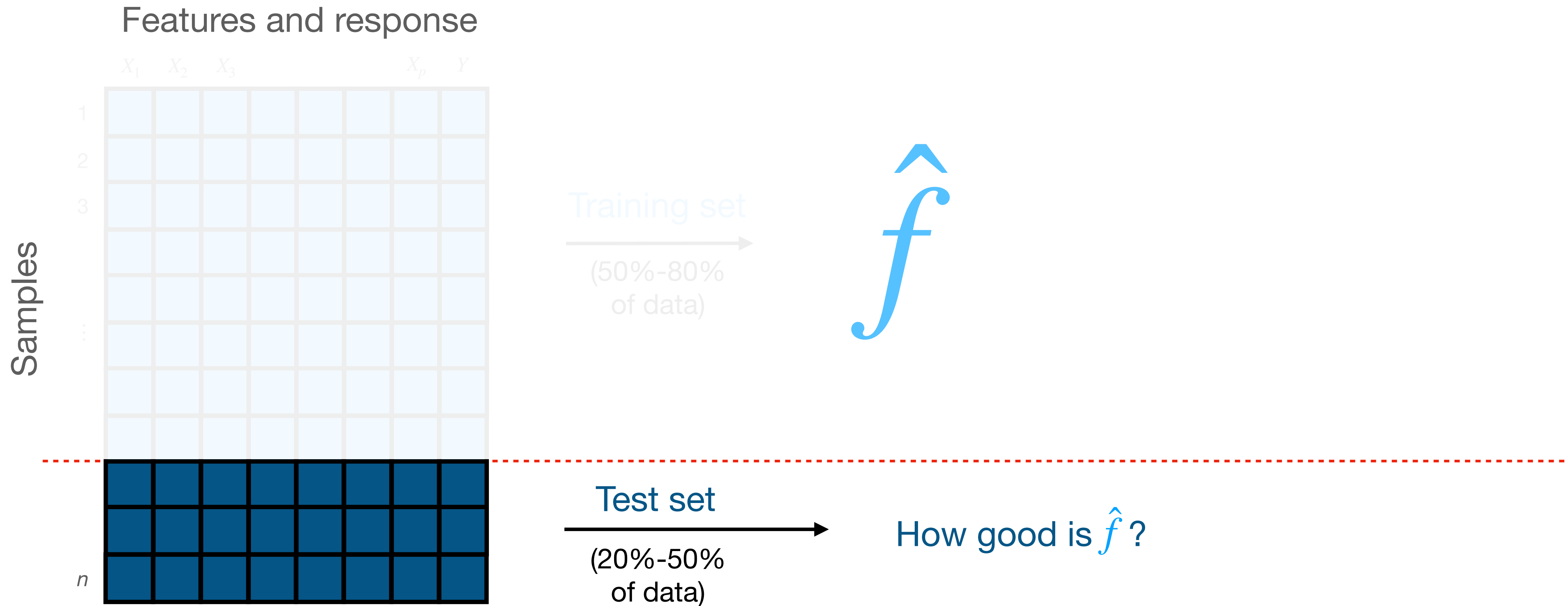
Separating data for model building and assessment



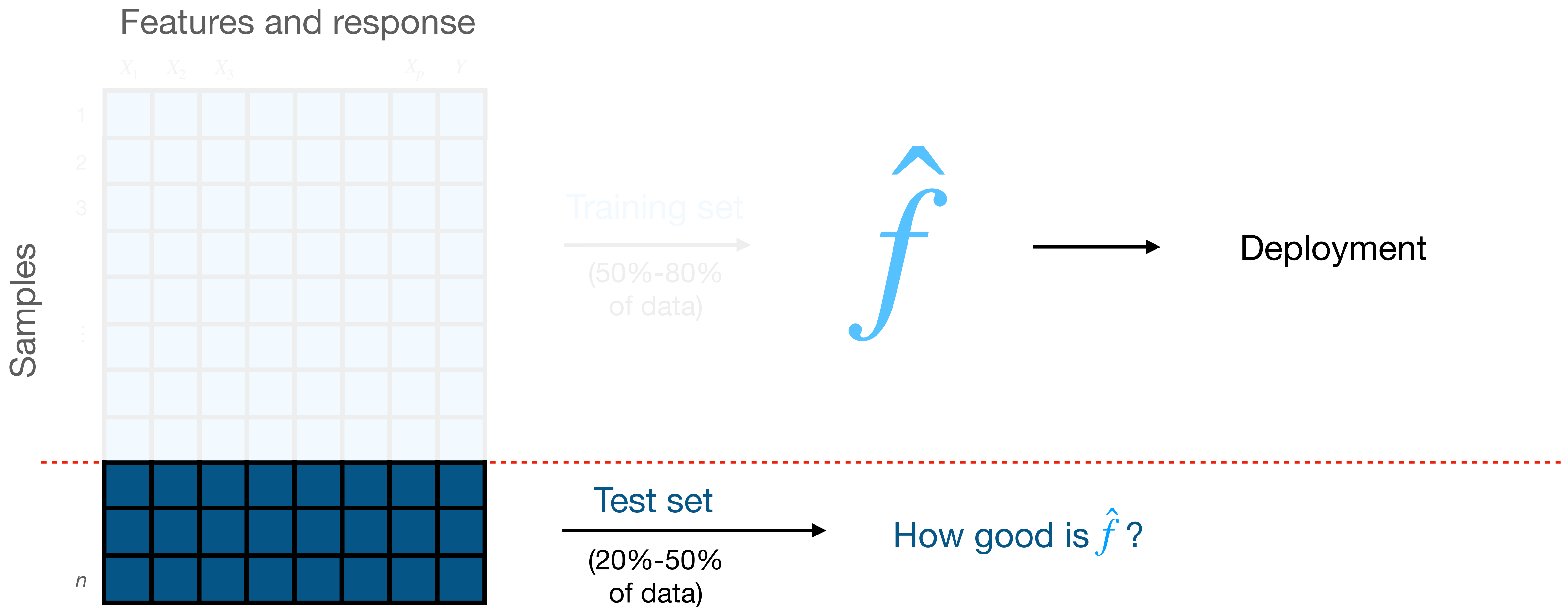
Separating data for model building and assessment



Separating data for model building and assessment



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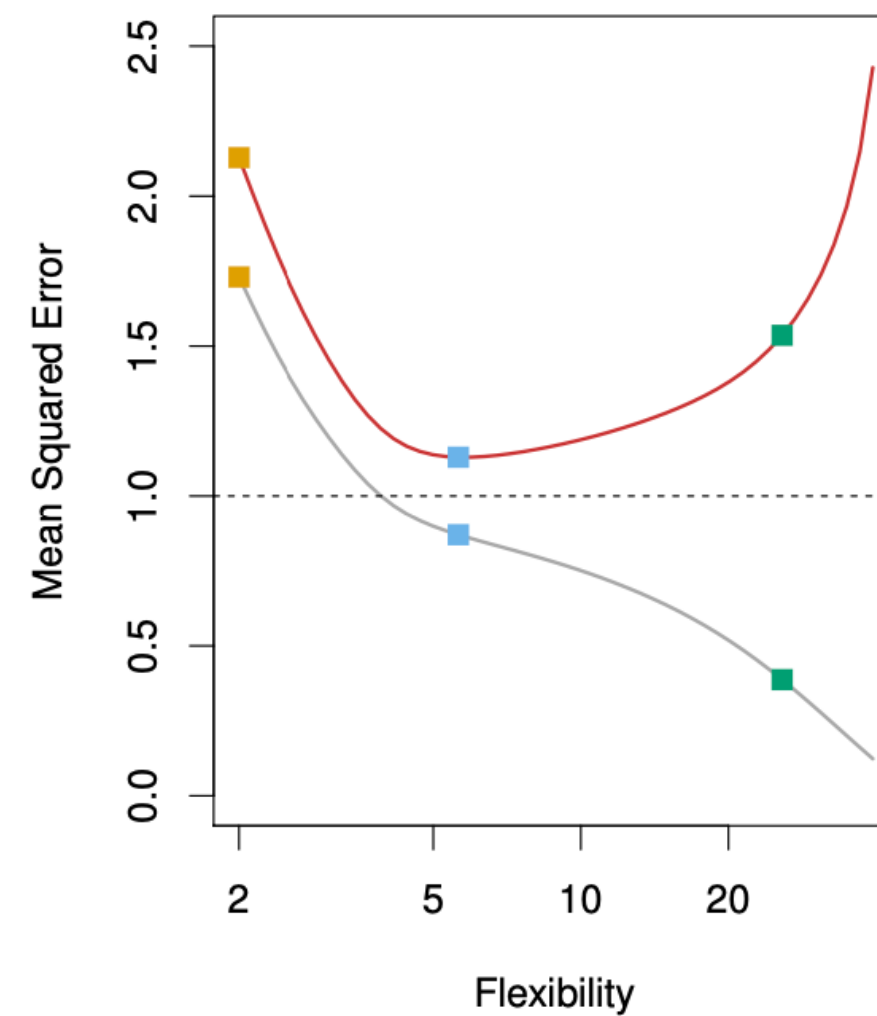
Mimicking test error on training set

Features and response

	X_1	X_2	X_3			X_p	Y
1							
2							
3							

Samples

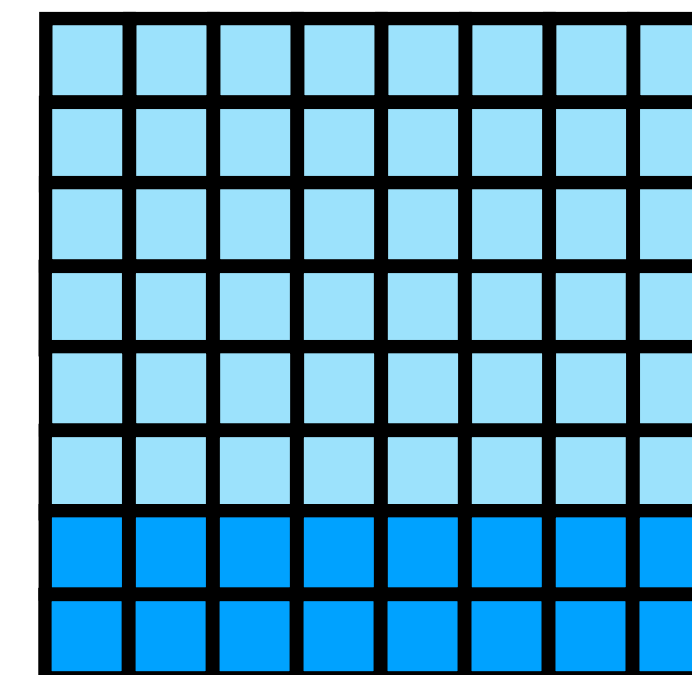
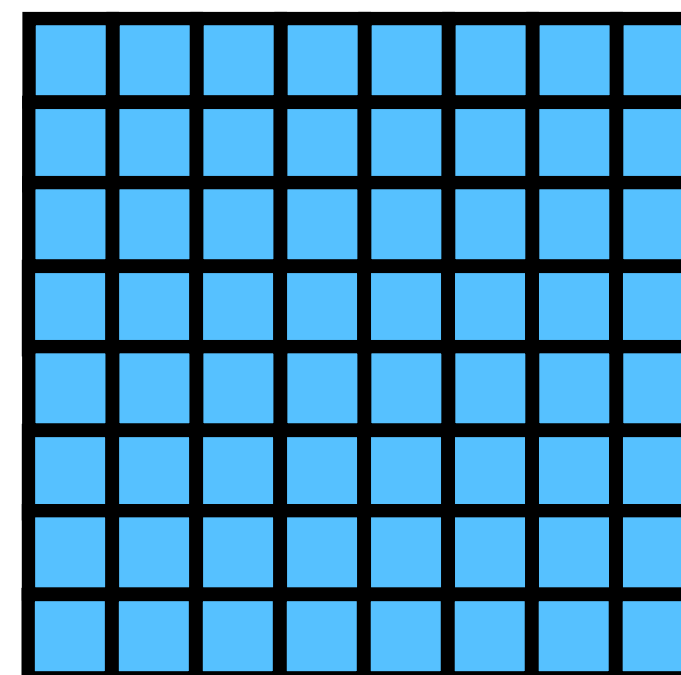
Training set
(50%-80%
of data)



Goal: find the lowest point on **test error curve**

Issue: test error is unknown (we don't see future samples)

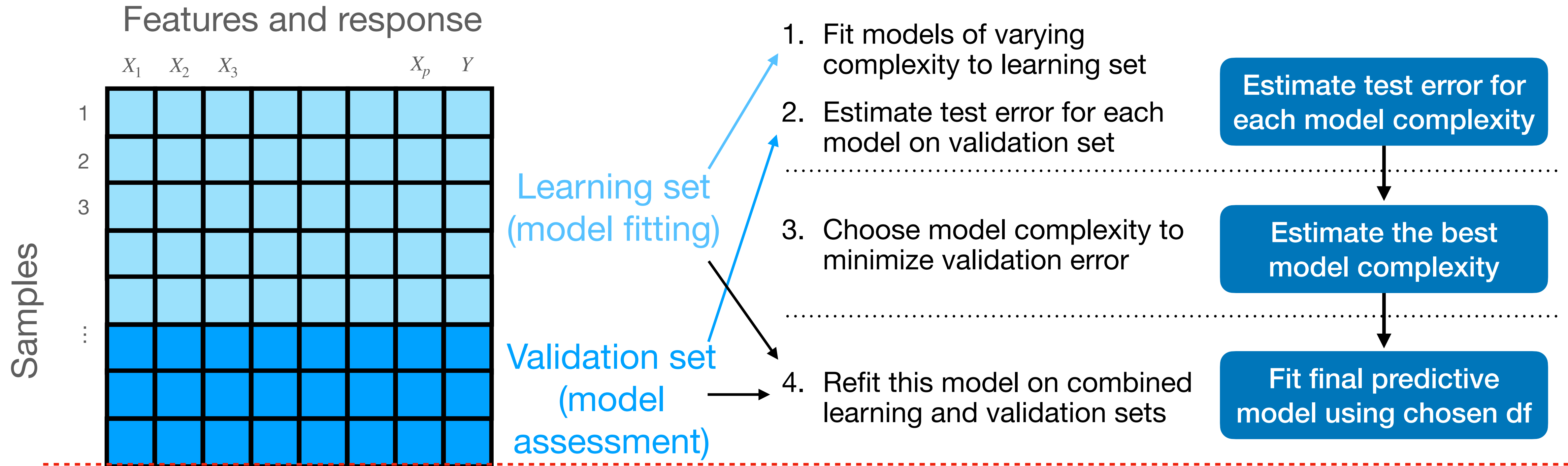
Solution: mimic test error using training set as all data



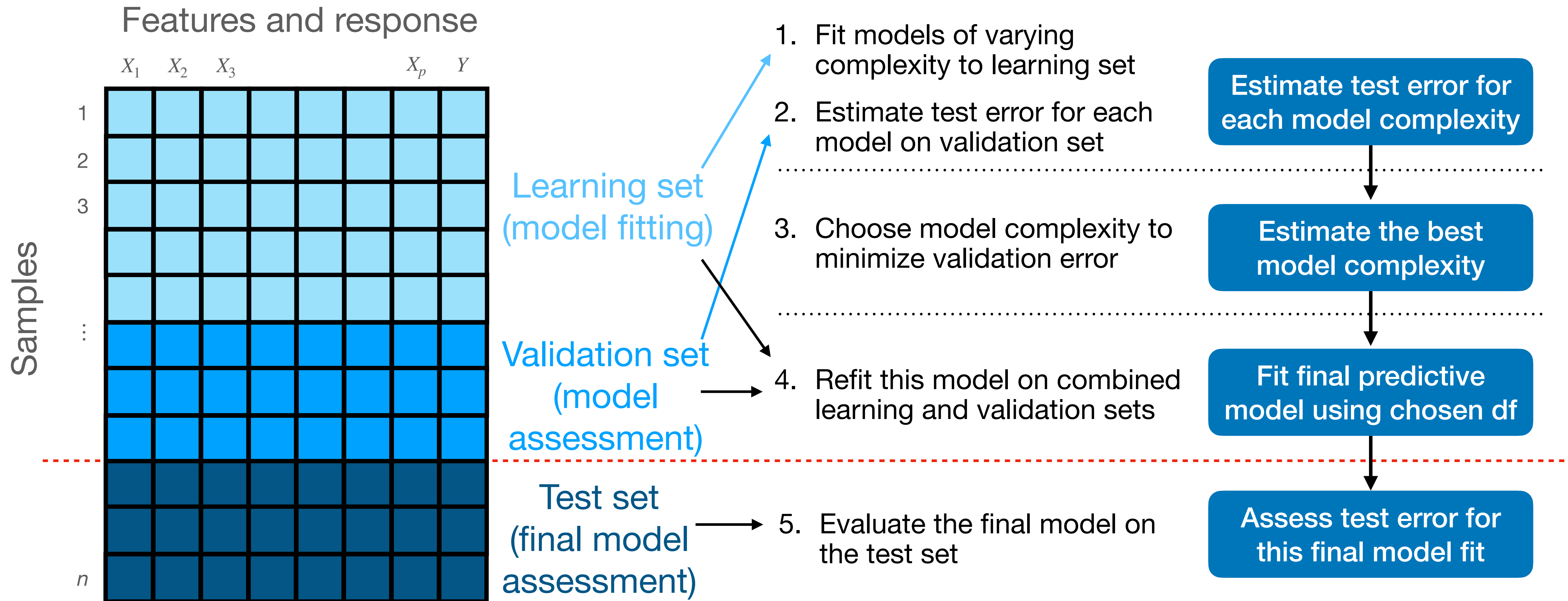
Learning set
(model fitting)

Validation set
(model assessment)

Validation set approach for model selection

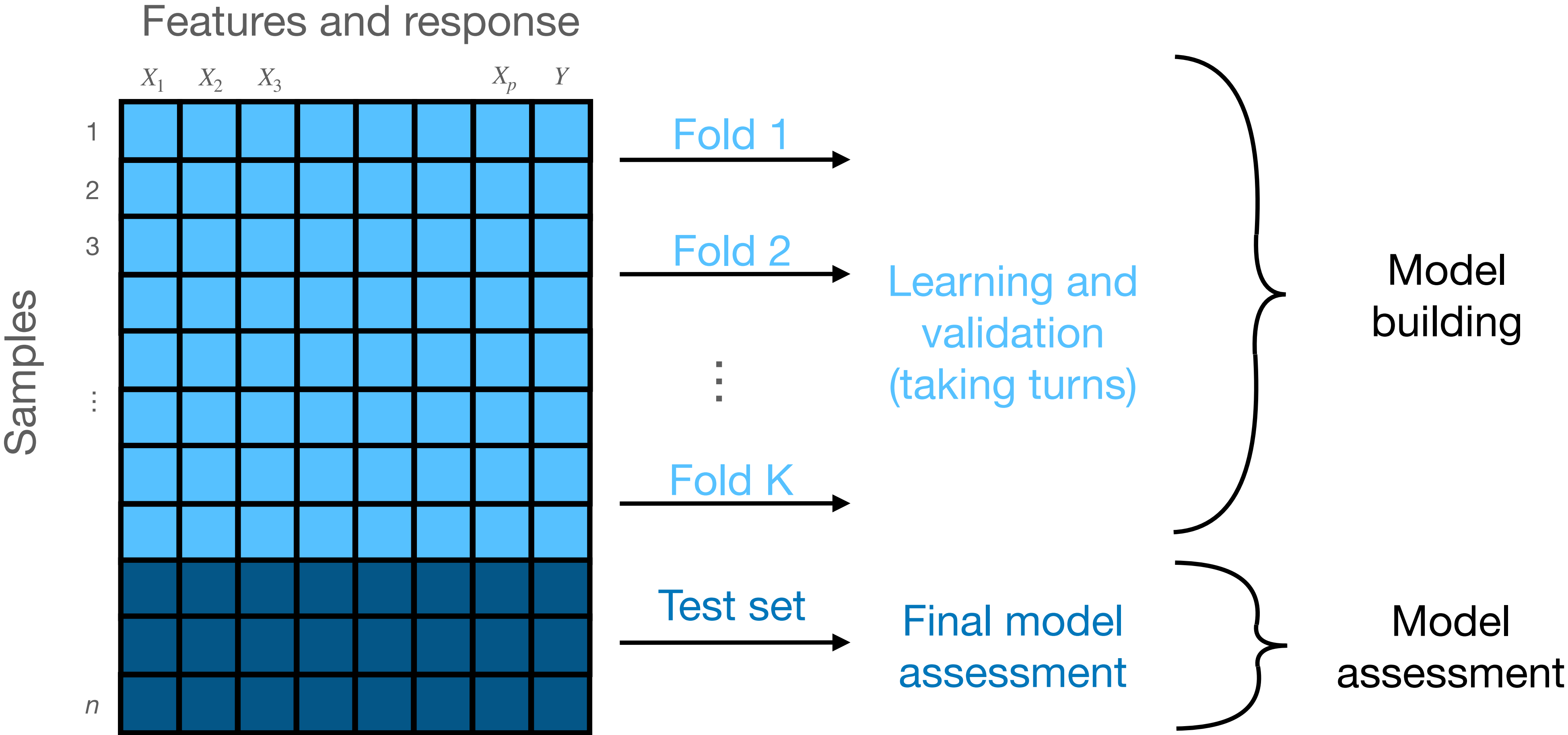


Validation set approach for model selection

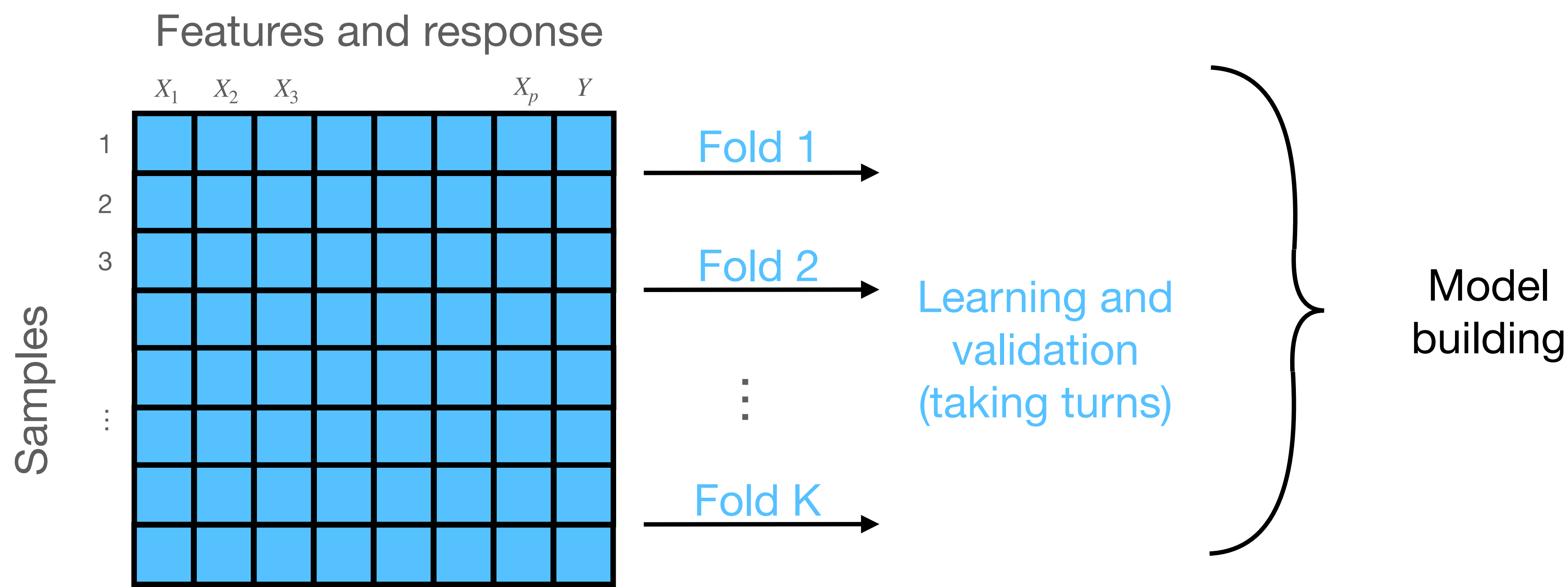


Drawback: Inefficient use of training samples,
e.g. small validation set may lead to poor model selection.

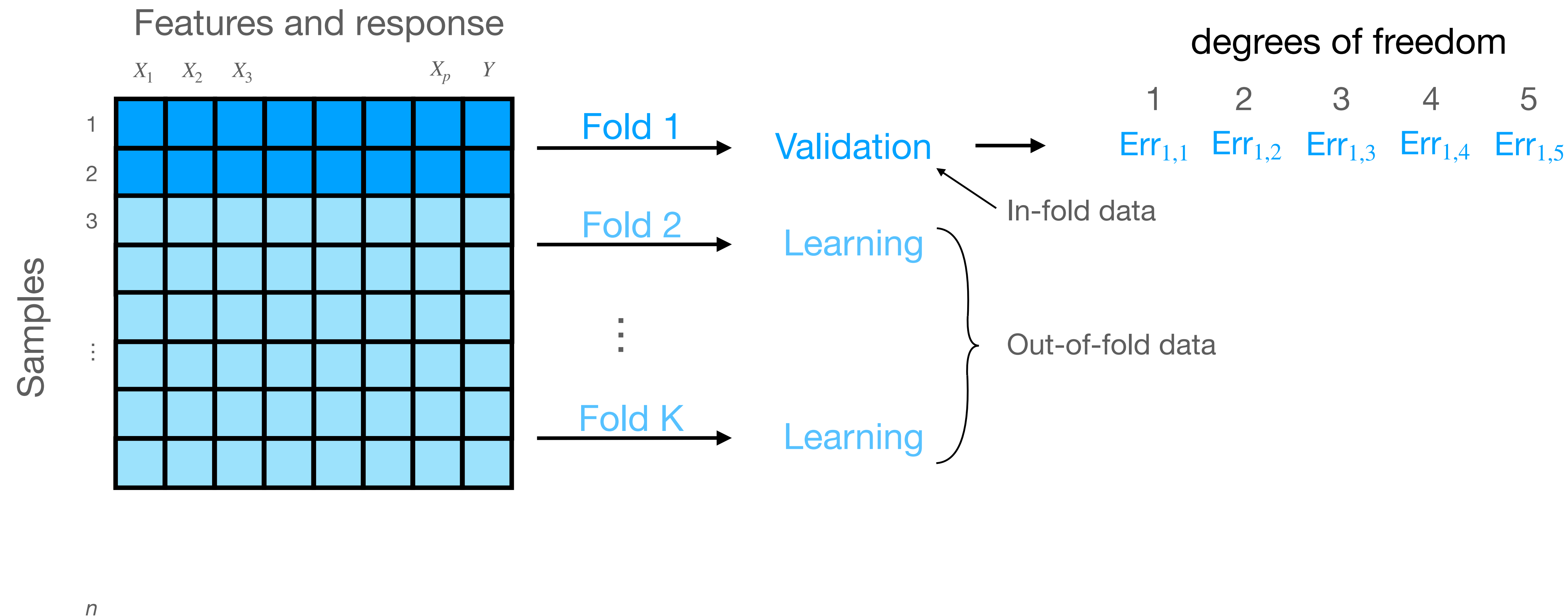
Cross-validation for model selection



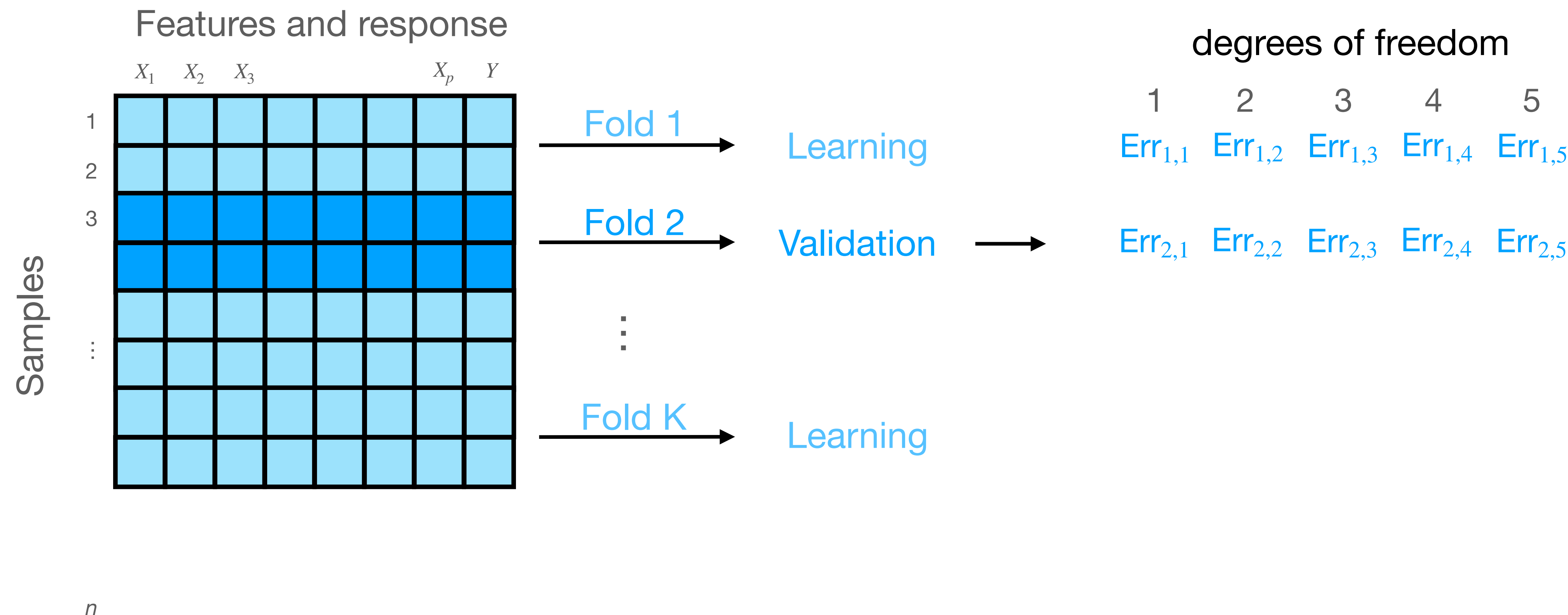
Cross-validation for model selection



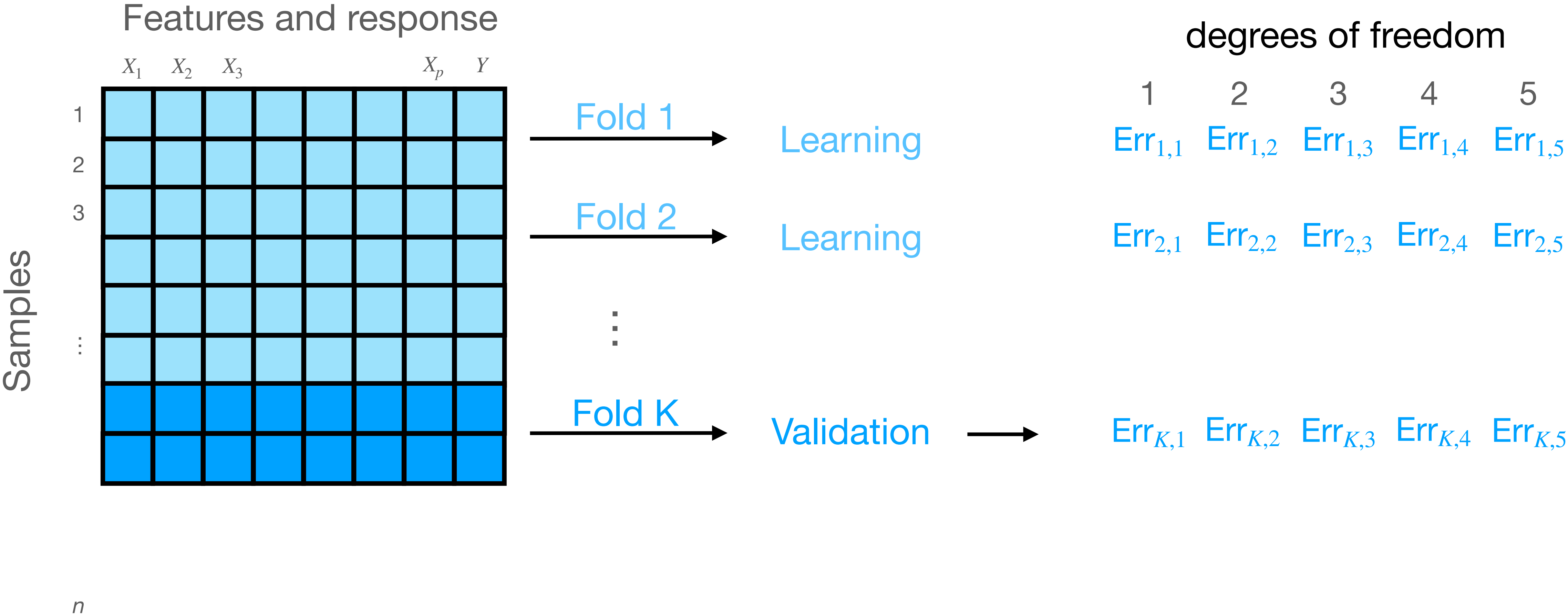
Cross-validation for model selection



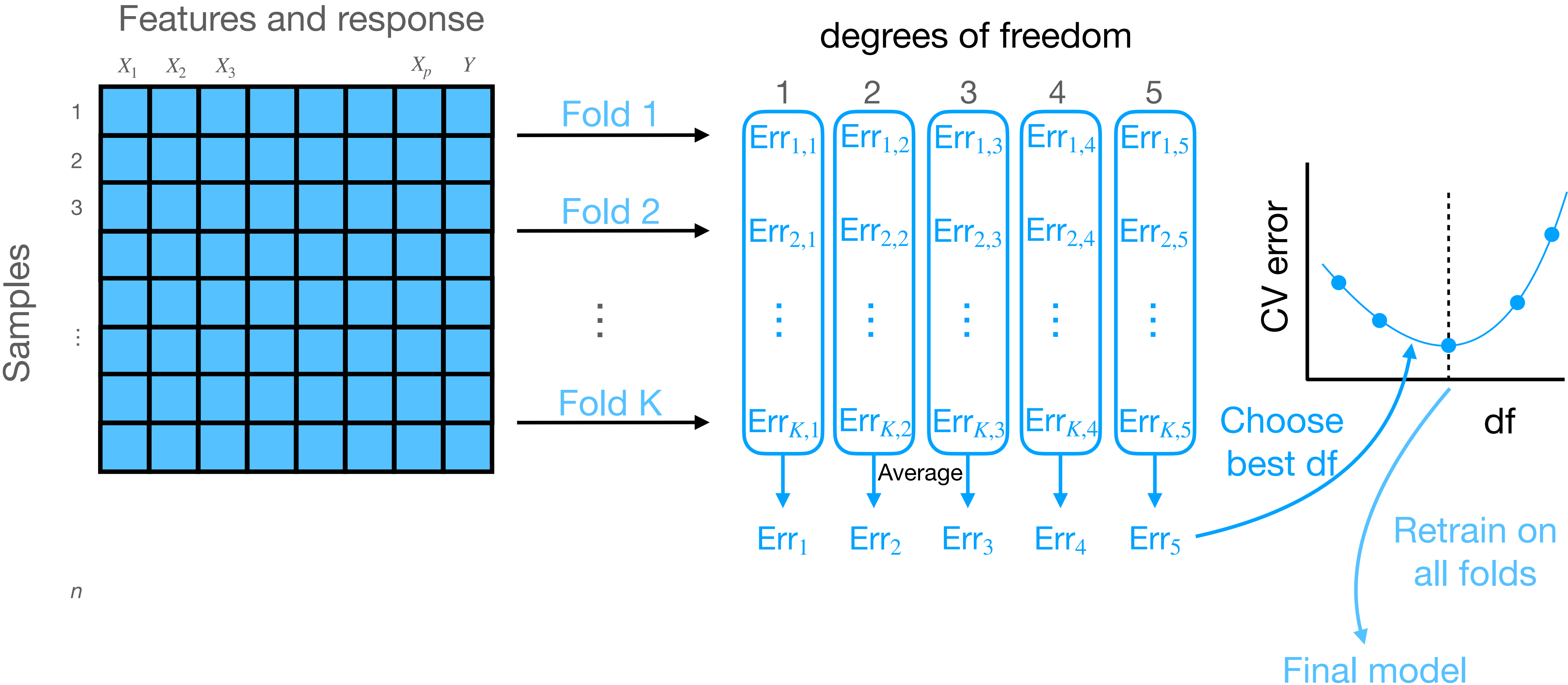
Cross-validation for model selection



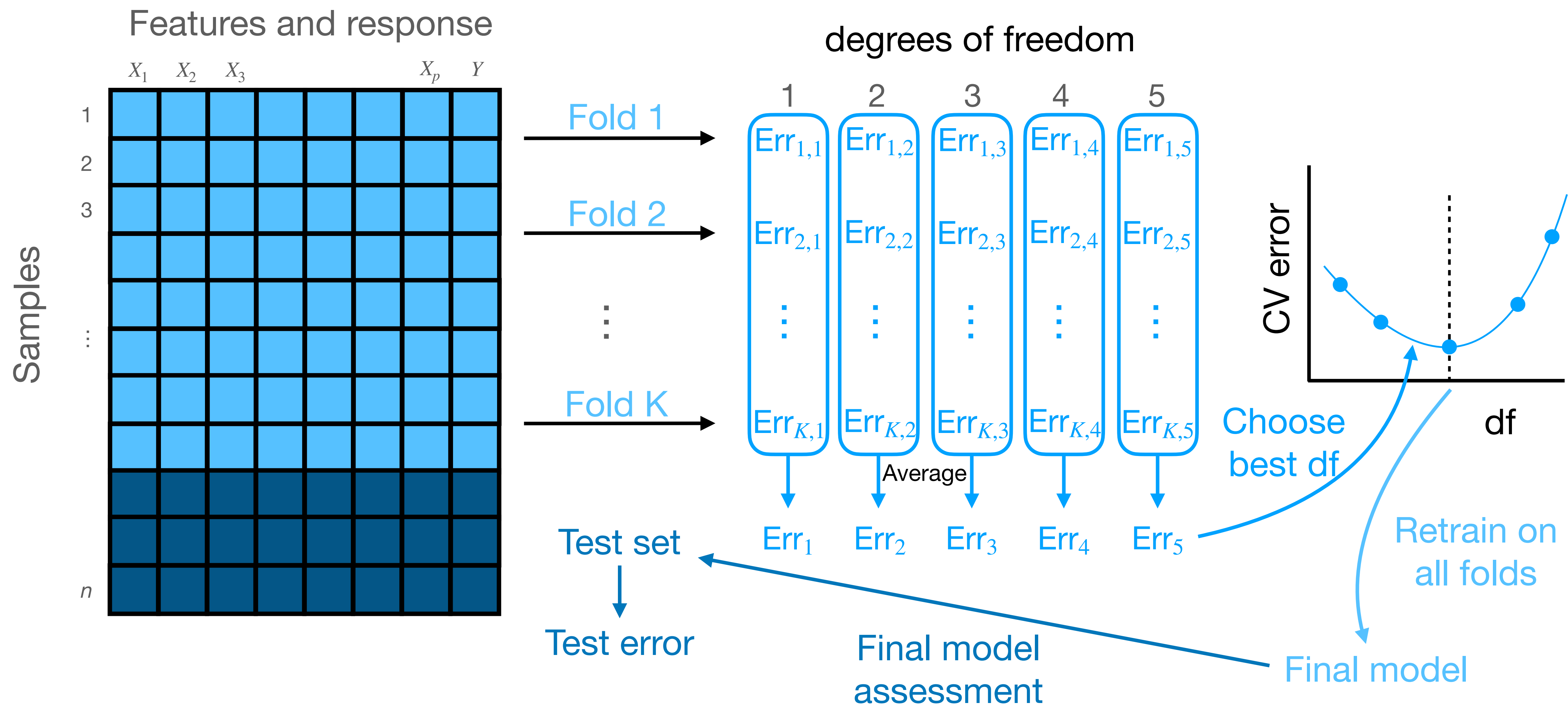
Cross-validation for model selection



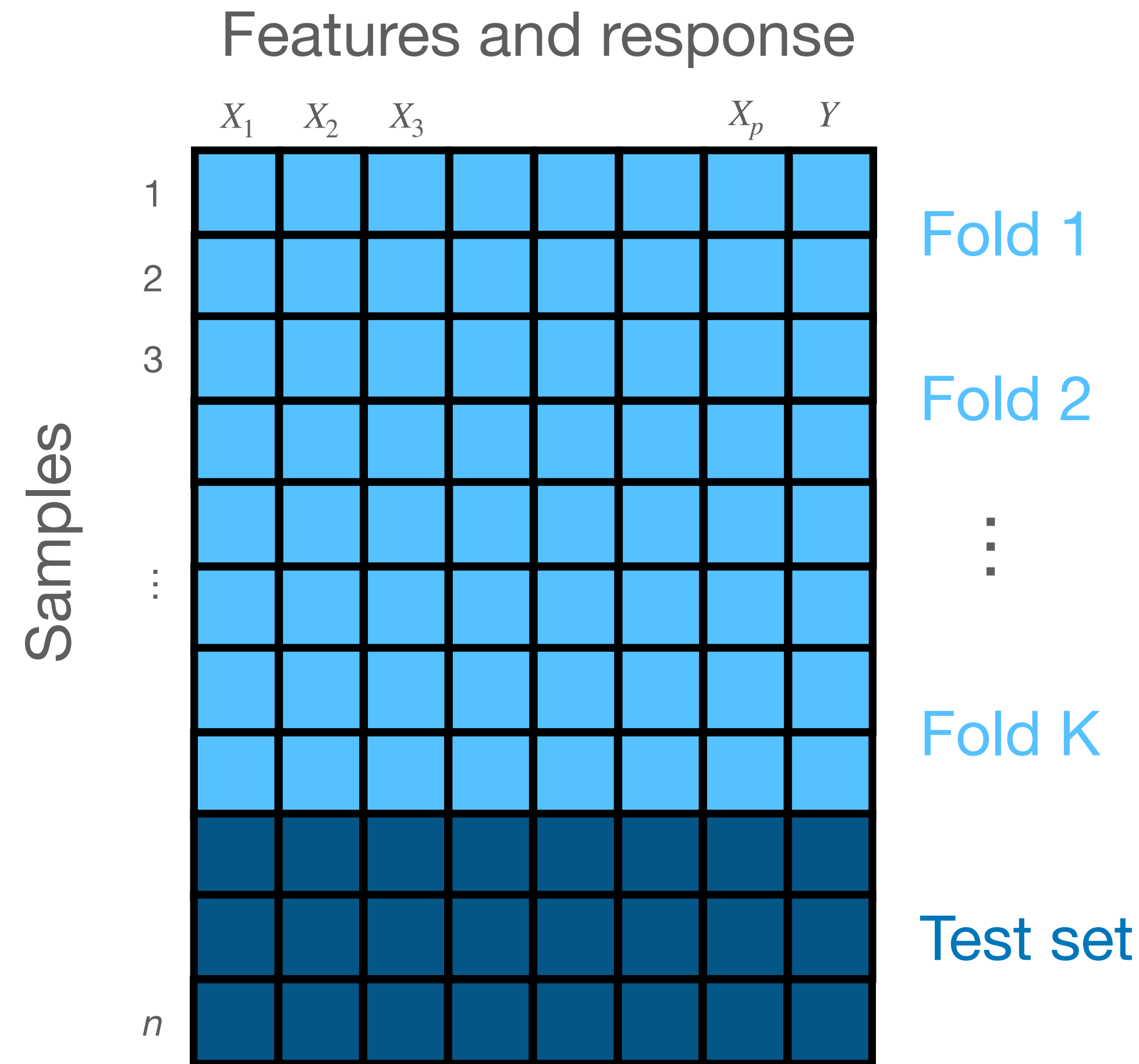
Cross-validation for model selection



Cross-validation for model selection



Cross-validation (summary)



1. Split training data into K folds
2. For each fold k ,
 - Fit models of varying complexity to training data, holding out fold k
 - Evaluate validation error for each model on fold k
3. Average across folds to get CV error for each model complexity
4. Choose model complexity to minimize CV error
5. Refit this model on all folds
6. Evaluate final model on the test set

Estimate test error for each model complexity

Estimate the best model complexity

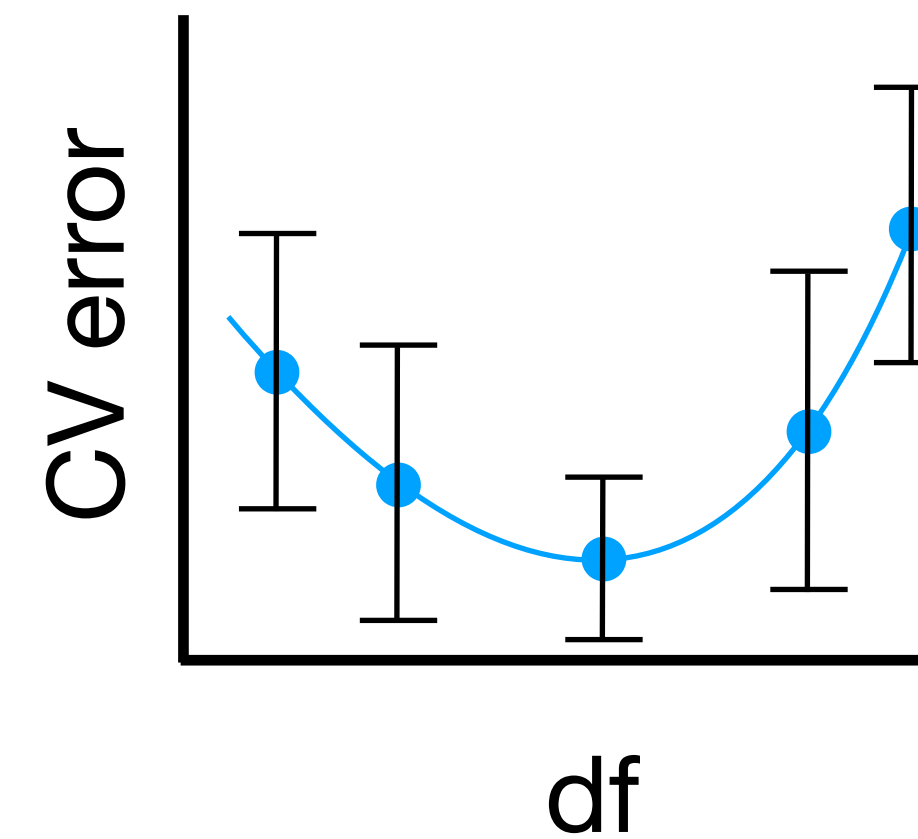
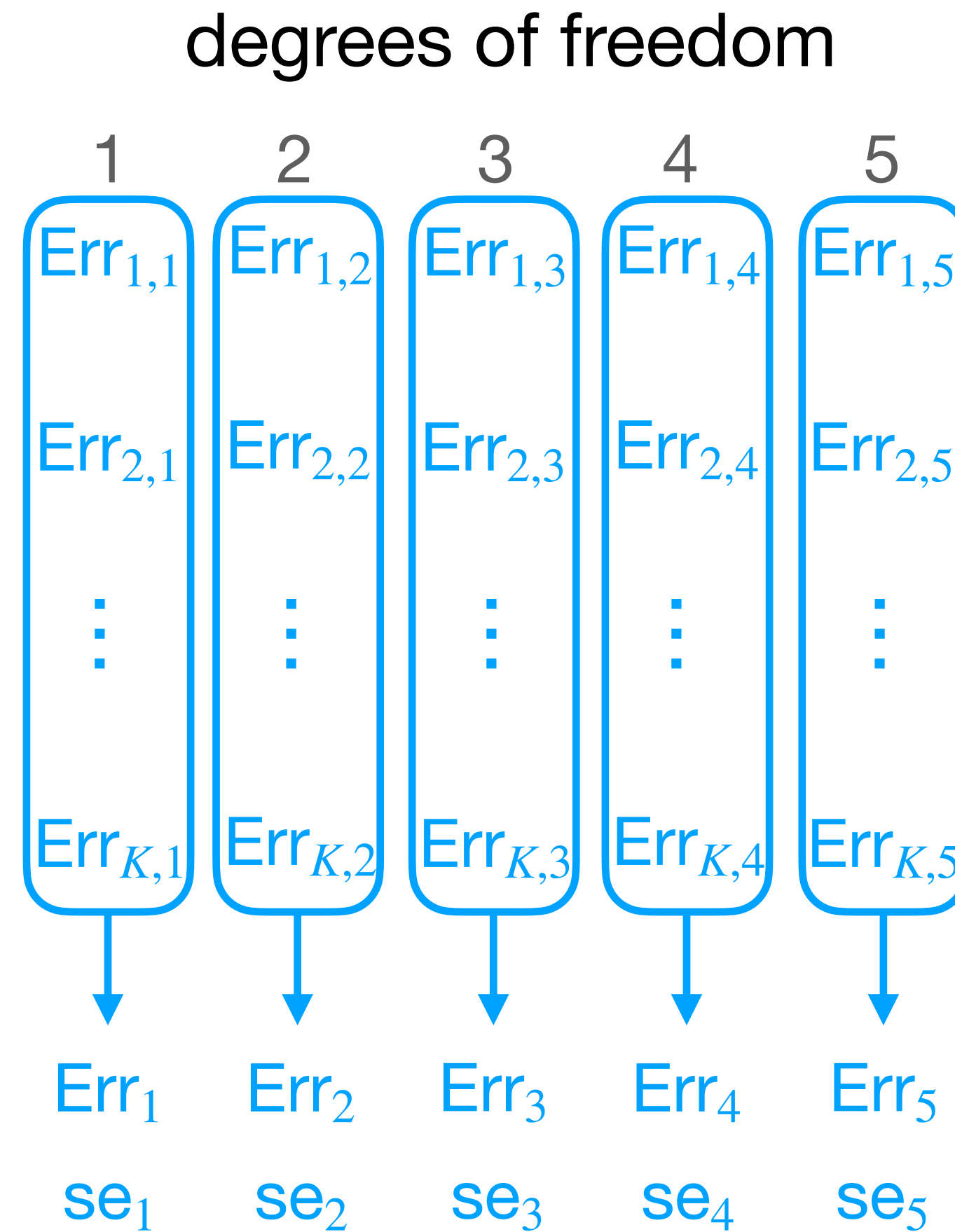
Fit final predictive model using chosen df

Assess test error for this final model fit

Choosing the number of folds

- More folds means more computation
- Fewer folds means the training sets used for model selection are much smaller than the actual training set
- (Advanced: More folds means that CV error estimates are more similar across folds, so the overall CV estimate defined as their average has more variance.)
- In practice, $K = 5$ or $K = 10$ are common choices

Cross-validation standard error

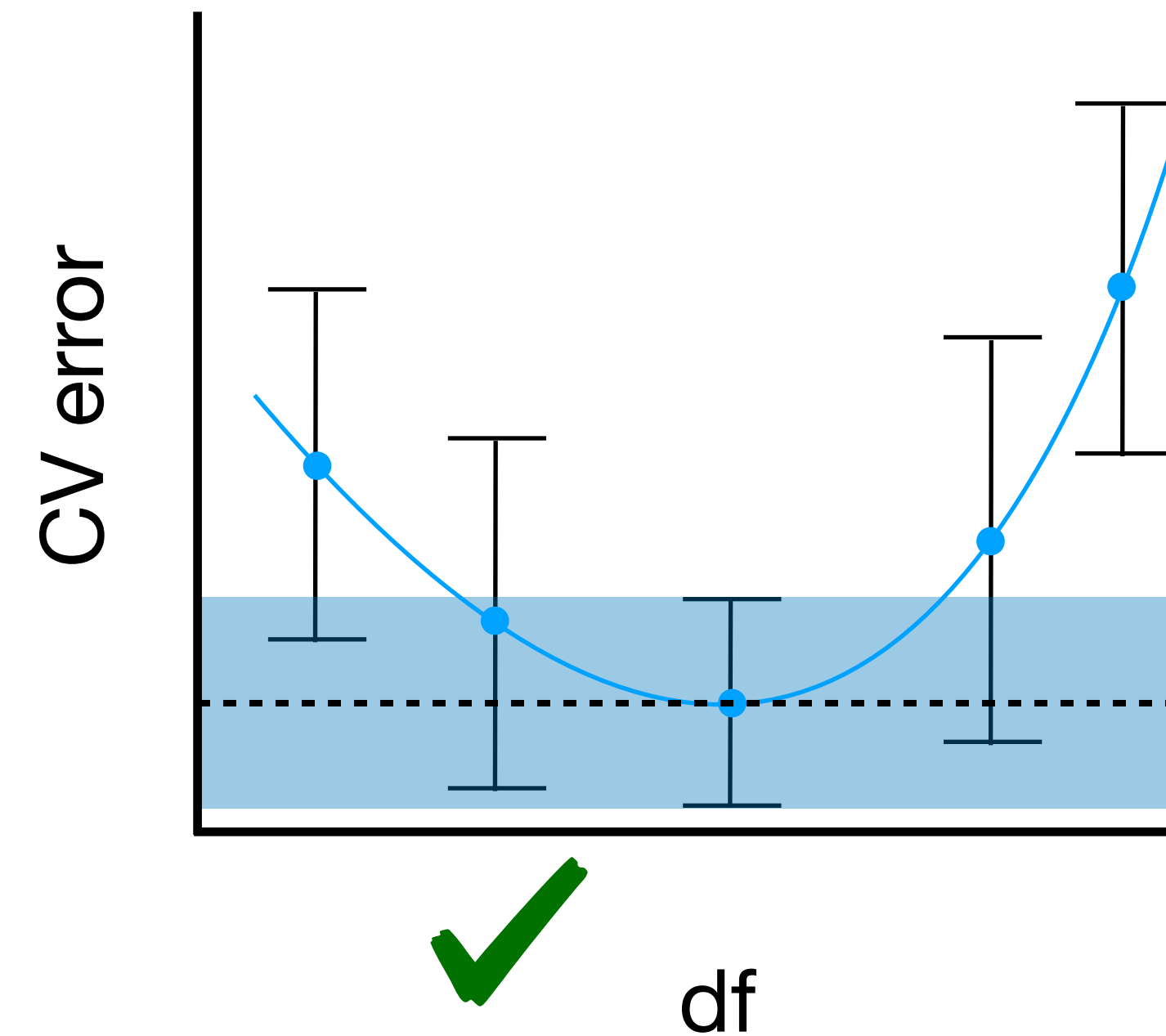


$$se_{df} = \frac{1}{\sqrt{K}} \times \text{s.d.}(Err_{1,df}, \dots, Err_{K,df})$$

One standard error rule

Occam's razor:

Select the smallest model for which the CV error is within one standard error of the lowest point on the curve.



Often used instead of choosing the minimum of CV curve.

Summary

